

Multi – level Linked List

–Print List(Main List and Child)

–Time Complexity Calculation using Recursive Tree Method

Program:

```
void printList(Node *node, int level = 0)
{
    while (node != nullptr)
    {
        for (int i = 0; i < level; i++)
            cout << "    ";
        cout << "|-- " << node->data << endl;

        if (node->child != nullptr)
        {
            printList(node->child, level + 1);
        }
        node = node->next;
    }
}

...

case 9:
    cout << "Current List Structure: " << endl;
    printList(head);
    break;
```

We have seen earlier :

$$2T\left(\frac{N-1}{2}\right) + C, \text{Base Case: } T(0) = 1$$

$$\Theta(N)$$

$\neq$ (NOT EQUAL TO)

$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2, \text{Base Case: } T(0) = 1.$$

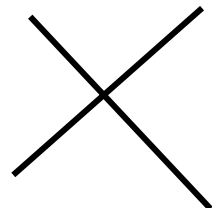
$$\Theta(N \log_2 N) \quad [\text{Correct}]$$

Which is Actual time complexity for Balance Tree of printlist .

Hence we cannot consider this generalized version :

$$2T\left(\frac{N-1}{2}\right) + C, \text{Base Case: } T(0) = 1$$

[Wrong]



$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2, \text{Base Case: } T(0) = 1.$$

$$\text{We know, } k = \log_2(n+1)$$

We see, c_1, c_2 and $+1$ are constant, can it be : $c \cdot \log_2 n$,

That is:

$$2T\left(\frac{N-1}{2}\right) + c \cdot \log_2 n$$

Recursion Step	No. of problems	Problem Size	Node's Level (k)	Work/Cost Per Problem ($c \cdot \log_2 n$)
0	1	n	0	$c \log_2(N)$
1	2	$\frac{n-1}{2}$	1	$c \times \log_2\left(\frac{N-1}{2}\right)$
2	4	$\frac{n-3}{4}$	2	$c \times \log_2\left(\frac{N-3}{4}\right)$
3	8	$\frac{n-7}{8}$	3	$c \log_2\left(\frac{N-7}{8}\right)$
.	.	.	.	.
.	.	.	.	.
k	2^k	$\frac{N - (2^k - 1)}{2^k}$	k	$c \log_2\left(\frac{N - 2^k + 1}{2^k}\right)$
$\log_2(N + 1)$ (Base Case)	N + 1	$\frac{N - (2^{\log_2(N+1)} - 1)}{2^{\log_2(N+1)}}$ = 0 (Base Case)	N/A	$T(0) = 1$

$$\frac{N - (2^{\log_2(N+1)} - 1)}{2^{\log_2(N+1)}} \text{ (Base Case)}$$

$$= \frac{N - (N + 1 - 1)}{N + 1}$$

$$= \frac{N - N - 1 + 1}{N + 1}$$

$$= \frac{0}{N+1} = 0$$

Also ,no. of problems : 2^k , when $k = \log_2(N+1) \Rightarrow 2^{\log_2(N+1)} \Rightarrow N+1$

Work done = Work per Problem × No. of Problems
$1 \times c \log_2(N) = c \log_2(N)$
$2 \left(c \times \log_2 \left(\frac{N-1}{2} \right) \right) = 2(c \times \log_2(N-1) - \log_2 2) = 2c(\log_2(N-1) - 1)$
$4 \left(c \times \log_2 \left(\frac{N-3}{4} \right) \right) = 4(c \times \log_2(N-3) - \log_2 2^2) = 4c(\log_2(N-3) - 2)$
$8 \left(c \times \log_2 \left(\frac{N-7}{8} \right) \right) = 8(c \times \log_2(N-7) - \log_2 2^3) = 8c(\log_2(N-7) - 3)$
$\vdots$
$2^k \left(c \times \log_2 \left(\frac{N - (2^k - 1)}{2^k} \right) \right) = 2^k(c \times \log_2(N - (2^k - 1)) - \log_2 2^k)$
$= 2^k(c \times \log_2(N - (2^k - 1)) - k)$
$(N+1) \times T(0)$

$$T(n) = 2c(\log_2(N-1) - 1) + 4c(\log_2(N-3) - 2) + 8c(\log_2(N-7) - 3) + \dots + 2^k(c \times \log_2(N - (2^k - 1)) - k) + (N+1) \times T(0)$$

$$T(n) = c \sum_{i=1}^k 2^i \left(\log_2 \left(N - (2^i - 1) \right) - i \right) + (N + 1) \times T(0)$$

$$T(n) = c \sum_{i=1}^k 2^i \left(\log_2 \left(N - (2^i - 1) \right) \right) - c \sum_{i=1}^k i 2^i + (N + 1) \times T(0)$$

We know from previous calculation:

$$\sum_{i=1}^k i 2^i = 2^{k+1}(k - 1) + 2$$

Hence,

$$T(n) = c \sum_{i=1}^k 2^i \left(\log_2 \left(N - (2^i - 1) \right) \right) - c(2^{k+1}(k - 1) + 2) \\ + (N + 1) \times T(0)$$

We already know from part 1:

$$N + 1 = 2^k$$

Lets ,go with 2nd Term:

$$\begin{aligned} & c(2^{k+1}(k-1) + 2) \\ &= c(2^k \times 2(k-1) + 2) \\ &= c((N+1) \times 2(k-1) + 2) [As, N+1 = 2^k] \\ &= c(2(N+1)(k-1) + 2) \end{aligned}$$

and,

$$k = \log_2(N+1)$$

Hence,

$$c(2(N+1)(\log_2(N+1) - 1) + 2)$$

Applying , Big Θ to the equation gives rise to :

$$= \Theta(N \log_2 N)$$

Hence,

$$\begin{aligned} T(n) &= c \sum_{i=1}^k 2^i (\log_2(N - (2^i - 1))) - \Theta(N \log_2 N) \\ &+ (N+1) \times T(0) \end{aligned}$$

Lets ,go with 1st Term:

$$c \sum_{i=1}^k 2^i (\log_2(N - (2^i - 1)))$$

Here,

$$N - (2^i - 1) = N - 2^i + 1 = (N + 1) - 2^i = 2^k - 2^i$$

Therefore we can say :

$$\log_2(N - (2^i - 1)) = \log_2(2^k - 2^i)$$

Factor out 2ⁱ :

$$2^k - 2^i = 2^{k-i+i} - 2^i = 2^i(2^{k-i} - 1)$$

Thus,

$$\log_2(2^k - 2^i) = \log_2 2^i(2^{k-i} - 1) = i + \log_2(2^{k-i} - 1)$$

$$\text{From: } [\log_2 ab = \log_2 a + \log_2 b]$$

∴ Substituting:

$$\begin{aligned}
c \sum_{i=1}^k 2^i (\log_2(N - (2^i - 1))) &= c \sum_{i=1}^k 2^i (i + \log_2(2^{k-i} - 1)) \\
&= c \sum_{i=1}^k i 2^i + c \sum_{i=1}^k 2^i (\log_2(2^{k-i} - 1)) \\
&= \Theta(N \log_2 N) + c \sum_{i=1}^k 2^i (\log_2(2^{k-i} - 1)) \left[c \sum_{i=1}^k i 2^i = \Theta(N \log_2 N) \right]
\end{aligned}$$

Hence , now we have :

$$\begin{aligned}
T(n) &= \Theta(N \log_2 N) + c \sum_{i=1}^k 2^i (\log_2(2^{k-i} - 1)) - \Theta(N \log_2 N) \\
&+ (N + 1) \times T(0)
\end{aligned}$$

Here we see:

$$\Theta(N \log_2 N) - \Theta(N \log_2 N) \left[\begin{array}{c} \text{This does not cancel out at} \\ \text{this form} \end{array} \right]$$

Lets put the actual value of $\Theta(N \log_2 N)$:

$$T(N) = c \sum_{i=1}^k i2^i + c \sum_{i=1}^k 2^i(\log_2(2^{k-i} - 1)) - c \sum_{i=1}^k i2^i + (N + 1) \times T(0)$$

Hence, $c \sum_{i=1}^k i2^i - c \sum_{i=1}^k i2^i$, cancels out.

∴ We are left with:

$$T(N) = c \sum_{i=1}^k 2^i(\log_2(2^{k-i} - 1)) + (N + 1) \times T(0)$$

To solve this final sum, we do the "bottom – up" variable swap.

A bottom – up variable swap is not a universally standard mathematical term, but in recurrence analysis and summations it usually means changing variables starting from the deepest level (the leaves) of the recursion tree and working upward.

Let j represent the distance from the bottom of the tree, so $j = k - i$.

Because $j = k - i$, it means $i = k - j$.

Let's substitute i out of our multiplier (2^i):

$$2^i = 2^{k-j} = \frac{2^k}{2^j}$$

Since we know: $2^k = N + 1$, we replace the top:

$$\frac{2^k}{2^j} = \frac{N+1}{2^j}$$

$$\Rightarrow 2^i(\log_2(2^{k-i} - 1)) = \frac{N+1}{2^j} \left(\log_2 \left(\frac{2^k}{2^i} - 1 \right) \right)$$

$$\Rightarrow 2^i(\log_2(2^{k-i} - 1)) = \frac{N+1}{2^j} \left(\log_2 \left(\frac{2^k}{\frac{1}{2^j}} - 1 \right) \right)$$

$$\Rightarrow 2^i(\log_2(2^{k-i} - 1)) = \frac{N+1}{2^j} \left(\log_2 \left(\frac{2^k}{1} \times \frac{2^j}{2^k} - 1 \right) \right)$$

$$\Rightarrow 2^i(\log_2(2^{k-i} - 1)) = \frac{N+1}{2^j} (\log_2(2^j - 1))$$

Now, plug j and our new fraction into the collapsed summation:

$$T(N) = c \sum_{j=1}^k \frac{N+1}{2^j} (\log_2(2^j - 1)) + (N+1) \times T(0)$$

$$T(N) = c(N+1) \sum_{j=1}^k \frac{1}{2^j} (\log_2(2^j - 1)) + (N+1) \times T(0)$$

$$T(N) = c(N+1) \sum_{j=1}^k \frac{(\log_2(2^j - 1))}{2^j} + (N+1) \times T(0)$$

Analyze the summation

For large j ,

$$2^j - 1 \approx 2^j$$

$$\text{so , } (\log_2(2^j - 1)) \approx \log_2 2^j = j$$

We know : $a - b = a \left(1 - \frac{b}{a}\right)$, applying this we get:

$$\text{Hence , } 2^j - 1 = 2^j \left(1 - \frac{1}{2^j}\right)$$

$$\therefore \log_2(2^j - 1) = \log_2 \left(2^j \left(1 - \frac{1}{2^j}\right)\right) = \log_2 2^j + \log_2 \left(1 - \frac{1}{2^j}\right) = j + \log_2 \left(1 - \frac{1}{2^j}\right)$$

The second term:

$$\log_2 \left(1 - \frac{1}{2^j}\right)$$

applying limit :

$$\lim_{j \rightarrow \infty} \log_2 \left(1 - \frac{1}{2^j}\right)$$

As $j \rightarrow \infty$,

$$2^j \rightarrow \infty$$

So,

$$\frac{1}{2^j} \rightarrow 0$$

$$\textit{Therefore , } 1 - \frac{1}{2^j} = 1 - 0 = 1$$

Since $\log_2(x)$ is continuous for $x > 0$,

$$\begin{aligned} & \lim_{j \rightarrow \infty} \log_2 \left(1 - \frac{1}{2^j} \right) \\ &= \log_2 \left(\lim_{j \rightarrow \infty} \left(1 - \frac{1}{2^j} \right) \right) \\ &= \log_2 \left(\lim_{j \rightarrow \infty} (1) - \lim_{j \rightarrow \infty} \frac{1}{2^j} \right) \\ &= \log_2 \left(1 - \frac{1}{\infty} \right) \\ &= \log_2 (1 - 0) \\ &= \log_2 (1) \\ &= 0 \end{aligned}$$

This is why, for large j ,

$$\log_2(2^j - 1) = j + \log_2 \left(1 - \frac{1}{2^j} \right) \approx j$$

hence we can write:

$$\log_2(2^j - 1) = j + \log_2\left(1 - \frac{1}{2^j}\right) = j + O(1) = j \left[as, \log_2\left(1 - \frac{1}{2^j}\right) = c \text{ or } O(1)\right]$$

and for asymptotic analysis,

$$\log_2(2^j - 1) = \Theta(j).$$

Thus your summand becomes:

$$\frac{\log_2(2^j - 1)}{2^j} \approx \Theta\left(\frac{j}{2^j}\right)$$

Hence,

$$\sum_{j=1}^k \frac{(\log_2(2^j - 1))}{2^j} \approx \Theta\left(\sum_{j=1}^k \frac{j}{2^j}\right)$$

Convergence – test

$$\sum_{j=1}^{\infty} \frac{j}{2^j}$$

if there exists an N so that for all $n \geq N$, $a_n \neq 0$ and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$

if $L < 1$, then $\sum a_n$ converges.

if $L > 1$, then $\sum a_n$ diverges.

if $L = 1$, then the test is inconclusive.

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{j+1}{2^{j+1}}}{\frac{j}{2^j}} \right|$$

$$\text{Divide fractions: } \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a \cdot d}{b \cdot c}$$

$$\left| \frac{\frac{j+1}{2^{j+1}}}{\frac{j}{2^j}} \right| = \left| \frac{(j+1) \times 2^j}{2^{j+1} \times j} \right|$$

$$\text{Applying exponent rule: } \frac{x^a}{x^b} = \frac{1}{x^{b-a}}$$

$$\left| \frac{(j+1) \times 2^j}{2^{j+1} \times j} \right| = \left| \frac{j+1}{2^{j-j+1} \times j} \right|$$

Add similar elements : $j + 1 - j = 1$

$$\left| \frac{j+1}{2 \times j} \right| = \left| \frac{j+1}{2j} \right|$$

Apply absolute rule : $|ax| = a|x|, a \geq 0$

$$|2j| = 2|j|$$

$$= \frac{|j+1|}{2|j|}$$

$$\lim_{j \rightarrow \infty} \left(\frac{|j+1|}{2|j|} \right)$$

$j + 1$ is positive when $j \rightarrow \infty$. Therefore $|j + 1| = j + 1$

$$= \lim_{j \rightarrow \infty} \left(\frac{j+1}{2|j|} \right)$$

j is positive when $j \rightarrow \infty$. Therefore $|j| = j$

$$= \lim_{j \rightarrow \infty} \left(\frac{j+1}{2j} \right)$$

We know $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$, applying we get:

$$= \frac{1}{2} \lim_{j \rightarrow \infty} \left(1 + \frac{1}{j} \right)$$

Applying $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$, we get:

$$= \frac{1}{2} \left(\lim_{j \rightarrow \infty} (1) + \lim_{j \rightarrow \infty} \left(\frac{1}{j} \right) \right)$$

$$= \frac{1}{2} \left(1 + \frac{1}{\infty} \right) \left[\lim_{x \rightarrow a} c = c, \text{ where } c \text{ is constant} \right]$$

$$= \frac{1}{2} (1 + 0) \left[\lim_{x \rightarrow \infty} \left(\frac{c}{x^a} \right) = \frac{c}{\infty} = 0 \right]$$

$$= \frac{1}{2} \times 1$$

$$= \frac{1}{2}$$

$L < 1$, by the ratio test.

$= \text{converges.}$

$$S = \sum_{j=1}^{\infty} \frac{j}{2^j} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$$

Multiplying $\frac{1}{2}$ to S , we get:

$$\frac{1}{2}S = \sum_{j=1}^{\infty} \frac{1}{2} \times \frac{j}{2^j} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots$$

Subtracting $\frac{1}{2}S$ from S , we get:

$$S - \frac{1}{2}S = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4}\right) + \left(\frac{3}{8} - \frac{2}{8}\right) + \left(\frac{4}{16} - \frac{3}{16}\right) + \dots$$

$$\frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\frac{1}{2}S = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots$$

Hence $r = \frac{1}{2}$ and $-1 < r < 1$, thus it satisfies the condition of infinite series which converge to a constant.

$$\text{Hence , } S_{\infty} = \frac{a}{1-r}$$

We can write

$$\frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\frac{1}{2}S = \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right)$$

$$\text{i.e. } S = a + ar + ar^2 + ar^3 + \dots$$

$$\text{Hence, from the series formula , we get: } a = \frac{1}{2}$$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

$$\Rightarrow \frac{1}{2}S_{\infty} = \frac{\frac{1}{2}}{1 - \frac{1}{2}}$$

$$\Rightarrow \frac{1}{2}S_{\infty} = \frac{1}{2} \times \frac{2}{1}$$

$$\Rightarrow S_{\infty} = 2$$

$$\therefore S = \sum_{j=1}^{\infty} \frac{j}{2^j} = 2$$

$$\text{Hence, } \sum_{j=1}^k \frac{(\log_2(2^j - 1))}{2^j} \approx \Theta \left(\sum_{j=1}^k \frac{j}{2^j} \right) = \Theta(c) \text{ or } \Theta(1)$$

$$\therefore T(N) = c(N + 1) \sum_{j=1}^k \frac{(\log_2(2^j - 1))}{2^j} + (N + 1) \times T(0)$$

$$\Rightarrow T(N) = c(N + 1) \times \Theta(1) + (N + 1) \times T(0)$$

We ,know , $T(0) = 1$, hence:

$$\Rightarrow T(N) = c(N + 1) \times \Theta(1) + (N + 1) \times 1$$

$$\Rightarrow T(N) = c(N + 1) \times \Theta(1) + (N + 1)$$

$$\Rightarrow T(N) = \Theta(N)$$

Hence we cannot consider this generalized version :

$$2T\left(\frac{N-1}{2}\right) + c \cdot \log_2 n, \text{ Base Case: } T(0) = 1$$

[Wrong]

Therefore ,it also implies ,we will strictly rely on:

$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2, \text{ Base Case: } T(0) = 1.$$

$$\Theta(N \log_2 N) \quad \text{[Correct]}$$
